

Assignment-1 : Chapter-1

Assignment Date: December 17th, 2025

Deadline: December 24th, 2025

Class: CSIT 5th Semester, New Summit College

Subject: Simulation & Modeling

Submission Mode: **Handwritten Document**

Questions:

1. Name the entities, attributes, activities, events, and state variables for the following systems:
 - a. Cafeteria
 - b. Inventory
 - c. Banking
 - d. A hospital emergency room
 - e. Communication
 - f. Gas station
 - g. Barber shop
2. Write short notes on:
 - a. System Boundary and System Environment
 - b. Real-time Simulation
 - c. PDE
3. What is a system? Explain different components of a system with a clear figure.
4. Why do we use differential equations in simulation?
5. Differentiate between discrete and continuous systems. **2079**
6. What is a model? What are the different types of models?

7. What is system modeling? What are the steps involved in system modeling? 2080
8. Differentiate between a dynamic physical model and a static physical model with an example. 2081
9. What do you understand by a static mathematical model? Explain with an example. Differentiate between stochastic and deterministic activities. 2076, 2082
10. Why is a model of a system built? What is a static model? Differentiate between static and dynamic mathematical models in simulation. 2080
11. What is simulation? Describe the analogy between a mechanical system and a corresponding electrical system with reference to the dynamic physical model. 2082
12. Differentiate between numerical and analytical methods?
13. Define simulation. Describe different phases of the simulation study with the help of a flowchart. 2081